

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] DECEMBER 2025 - JANUARY 2026

Paper Code: ES-119**Subject: Manufacturing Processes****Time: 3 Hours****Maximum Marks: 60****Note: Attempt all questions as directed. Internal choice is indicated.**

Q. No.	Question	Marks
Q.1	Attempt any four of the following questions:	(4 × 5 = 20)
(a)	Define Manufacturing. Explain its fundamental importance towards technological and socio-economic development.	5
(b)	What is Riser in a casting mould? Explain why it is used and how it prevents shrinkage voids.	5
(c)	Define any five mechanical properties of engineering metals with examples.	5
(d)	What do you mean by Direct Current Straight Polarity (DCSP) and Direct Current Reverse Polarity (DCRP)? Explain heat distribution.	5
(e)	Why is edge preparation needed in welding operations? Under what thickness is it mandatory?	5
(f)	Explain the major characteristics and diverse engineering applications of commercial plastics.	5
(g)	Differentiate between hot working and cold working processes of metals in detail.	5
(h)	How do thermosetting plastics differ from thermoplastics? Give distinct molecular structures and examples.	5
Q.2	Enlist the different types of patterns used in foundries. Discuss any four in detail with neat sketches.	5
(a)		
(b)	Explain shell mould casting with a neat labeled sketch; state its advantages, limitations, and applications.	5
	OR	
Q.3	Explain the construction, operation, and various metallurgical zones of a Cupola furnace with a neat labeled sketch.	10

Q. No.	Question	Marks
Q.4 (a)	Explain the different types of flames in oxy-acetylene gas welding with neat sketches and applications.	6
(b)	Classify welding electrodes along with their specific functions and coating ingredients.	4
	OR	
Q.5 (a)	Explain with a diagram the working principle, advantages, limitations, and applications of Gas Metal Arc Welding (GMAW).	6
(b)	Explain any four welding defects, their metallurgical causes, and preventive remedies.	4
Q.6 (a)	Explain the working principle of extrusion. Differentiate between hot and cold extrusion processes.	6
(b)	Explain the following sheet metal operations with diagrams: (i) Shearing (ii) Bending (iii) Deep Drawing (iv) Metal Spinning.	4
	OR	
Q.7 (a)	Define plastic deformation process. Explain the principle of metal rolling in detail with a schematic diagram.	6
(b)	Differentiate between wire drawing and rod drawing operations.	4
Q.8	Define powder metallurgy. Explain the sequential stages of the powder metallurgy process in detail.	10
	OR	
Q.9	Explain the following polymer processing techniques with neat sketches: (i) Injection moulding (ii) Blow moulding (iii) Rotational moulding.	10

**** End of Question Paper ****